DESCENDING HOMOMORPHISMS BETWEEN DERIVED OBJECTS ALONG A RELATIVE LIMIT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12451-12496, immediately following the statement at 12434-12449.

Editorial note. Seven cross-chapter pointers (Derived Categories of Spaces/Schemes) plus the chapter's own situation are recorded in assembly.external_references.

Editorial note. The colimit comparison map is displayed verbatim in the statement span.

Allowed standard prerequisites (editorial).

- The situation of relative descent. chapter situation situation-relative-descent, cited in the statement. Fixes the limit presentation X = lim X_i over X_0 and the pullbacks K_i, L_i whose Hom sets are compared.
- The schemes-level descent statement. Derived Categories of Schemes, lemma perfect-lemma-relative-descend-homomorphisms, cited in the proof. Supplies the argument in the scheme case, which is transferred to algebraic spaces here.
- Induction principle and Mayer-Vietoris for derived categories of spaces. Derived Categories of Spaces, lemmas spaces-perfect-lemma-induction-principle and spaces-perfect-lemma-mayer-vietoris-hom, cited in the proof. Reduce the verification to affine pieces where the comparison can be made.
- Descending pseudo-coherence and tor dimension. Derived Categories of Spaces, lemmas spaces-perfect-lemma-descend-pseudo-coherent and spaces-perfect-lemma-tor-dimension-rel, cited in the proof. Keep the relevant finiteness conditions along the descent.
- The small etale site comparison. Derived Categories of Spaces, lemma spaces-perfect-lemma-derived-quasi-coherent-small-etale-site, cited in the proof. Moves the computation between the derived category of the space and that of its small etale site.

Source reference. Original reference 0DI7: focused-seeds/geometry/S9-spaces-relative-descend-homomorphisms-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DKU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12421–12421. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 071Q: focused-seeds/geometry/S9-spaces-relative-descend-homomorphisms-spaces/ref-tags.txt, source lines 6–6. Navigation pointer

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

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Source reference. Original reference 08HE: focused-seeds/geometry/S9-spaces-relative-descend-homomorphisms-spaces/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08GP: focused-seeds/geometry/S9-spaces-relative-descend-homomorphisms-spaces/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08H9: focused-seeds/geometry/S9-spaces-relative-descend-homomorphisms-spaces/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08JL: focused-seeds/geometry/S9-spaces-relative-descend-homomorphisms-spaces/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DK7: focused-seeds/geometry/S9-spaces-relative-descend-homomorphisms-spaces/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0DKU. Let K_0 and L_0 be objects of $D(\mathcal{O}_{X_0})$. Set $K_i = Lf_{i0}^*K_0$ and $L_i = Lf_{i0}^*L_0$ for $i \geq 0$ and set $K = Lf_0^*K_0$ and $L = Lf_0^*L_0$. Then the map*

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(\mathcal{O}_{X_i})}(K_i, L_i) \longrightarrow \operatorname{Hom}_{D(\mathcal{O}_X)}(K, L)$$

is an isomorphism if K_0 is pseudo-coherent and $L_0 \in D_{QCoh}(\mathcal{O}_{X_0})$ has (locally) finite tor dimension as an object of $D((X_0 \to Y_0)^{-1}\mathcal{O}_{Y_0})$

Proof. For every quasi-compact and quasi-separated object U_0 of $(X_0)_{spaces,étale}$ consider the condition P that

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(\mathcal{O}_{U_i})}(K_i|_{U_i}, L_i|_{U_i}) \longrightarrow \operatorname{Hom}_{D(\mathcal{O}_U)}(K|_U, L|_U)$$

is an isomorphism where $U = X \times_{X_0} U_0$ and $U_i = X_i \times_{X_0} U_0$. We will prove P holds for each U_0.

Suppose that $(U_0 \subset W_0, V_0 \to W_0)$ is an elementary distinguished square in $(X_0)_{spaces,étale}$ and P holds for $U_0, V_0, U_0 \times_{W_0} V_0$. Then P holds for W_0 by Mayer-Vietoris for hom in the derived category, see Derived Categories of Spaces, Lemma 08H9.

We first consider $U_0 = W_0 \times_{Y_0} X_0$ with W_0 a quasi-compact and quasi-separated object of $(Y_0)_{spaces,étale}$. By the induction principle of Derived Categories of Spaces, Lemma 08GP applied to these W_0 and the previous paragraph, we find that it is enough to prove P for $U_0 = W_0 \times_{Y_0} X_0$ with W_0 affine. In other words, we have

reduced to the case where Y_0 is affine. Next, we apply the induction principle again, this time to all quasi-compact and quasi-separated opens of X_0, to reduce to the case where X_0 is affine as well.

If X_0 and Y_0 are affine, then we are back in the case of schemes which is proved in Derived Categories of Schemes, Lemma 0DI7. The reader may use Derived Categories of Spaces, Lemmas 08JL, 071Q, 08HE, and 0DK7 to accomplish the translation of the statement into a statement involving only schemes and derived categories of modules on schemes. □